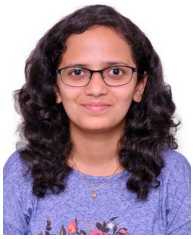
PRIYANKA GAWADE



Machine Learning Researcher | Scientific AI | Statistical Inference | Physical Modelling

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PROFESSIONAL PROFILE

Machine-learning researcher with a PhD in astrophysics and a B.Tech in Mechanical Engineering, currently a postdoctoral researcher at the University of Rome Tor Vergata. Expertise in physical modelling, statistical inference, uncertainty quantification, covariance estimation, and scientific data analysis. Designed and implemented Convolutional Neural Network (CNN) and Bayesian Neural Network (BNN) architectures for galaxy-scale strong-lens inference; also explored autoencoder-based denoising. Proficient in Python, TensorFlow, and Keras, with experience in multi-channel astronomical imaging and ML-based transient detection and classification in Rubin LSST data.

CORE EXPERTISE RELEVANT TO SINTEF ANALYTICS & AI

Scientific AI & ML	CNNs, BNNs, autoencoder-based denoising; predictive modelling and deep learning
Statistical Inference	Parameter inference, uncertainty quantification, covariance estimation, probabilistic modelling
Physical-System Modelling	Galaxy-scale strong-lensing models; simulation-driven inference and validation on real data
Multi-Channel Scientific Data	Multi-channel astronomical imaging; transient survey data
Programming & Tools	Python, TensorFlow, Keras, Linux, LaTeX
Research & Professional Strengths	Complex problem solving, teamwork, scientific communication, strong English proficiency, reliability, and inclusive research values

RESEARCH & MACHINE LEARNING EXPERIENCE

Postdoctoral Researcher — University of Rome Tor Vergata, Rome, Italy **2025–Present**

- Developing machine-learning algorithms for detection and classification of transient astronomical sources in Rubin LSST data.
- Contributing to collaborative research connected with Rubin LSST and high-energy transient science.

PhD Researcher — IUCAA, Pune, India **2020–2025**

- Designed and implemented CNN architectures to infer physical parameters of galaxy-scale strong gravitational lenses from Hyper Suprime-Cam imaging data.
- Developed BNN models to estimate lens and source parameters together with their uncertainties and covariances.
- Used multi-channel astronomical images as inputs for physical parameter inference.
- Built simulated strong-lensing datasets, validated models on real observations, and investigated simulation-to-real discrepancies.
- Explored autoencoder-based denoising methods for astronomical imaging data.
- Investigated gravitational-lensing probes of dark matter, including primordial-black-hole constraints using GRB microlensing parallax.

EDUCATION

PhD, Astrophysics — IUCAA, Pune, India **2020–2025**
Graduate School — IUCAA, Pune, India **2019–2020**
Integrated MSc-PhD — IISER Pune, Pune, India **2017–2019**
B.Tech, Mechanical Engineering — VIT Pune, Pune, India **2012–2016**

SELECTED PUBLICATIONS

- **Gawade et al. (2026):** Bayesian neural network prediction of lens and source parameters for strong lensing samples from Hyper Suprime-Cam Survey. Under review in MNRAS.
- **Gawade et al. (2025):** Neural network prediction of model parameters for strong lensing samples from Hyper Suprime-Cam Survey. MNRAS. DOI: 10.1093/mnras/staf935.

- **Gawade et al. (2023):** On the feasibility of primordial black hole abundance constraints using lensing parallax of GRBs. MNRAS. DOI: 10.1093/mnras/stad3336.
- **Bhalerao, Sawant, ..., Gawade et al. (2024):** Science with the Daksha High Energy Transients Mission. Experimental Astronomy 57, 23. DOI: 10.1007/s10686-024-09923-1.

COLLABORATIONS & SCIENTIFIC COMMUNICATION

Rubin LSST | Daksha Science

Sep 2025 Bayesian neural network prediction of lens and source parameters along with their uncertainties for strong lenses from HSC — LSST@Europe7.

Mar 2024 Neural network prediction of model parameters of strong lenses from HSC — Rubin LSST Strong Lensing Meeting, University of Oxford, UK.

Mar 2024 PBH dark-matter constraints using microlensing parallax of gamma-ray bursts — Multi-messenger Gravitational Lensing, Royal Society Meeting, Manchester, UK.

TEACHING, SCIENCE OUTREACH & LANGUAGES

- Assisted with tutorials and grading for an MSc-level Cosmology course taught by Dr. Surhud More, University of Pune (Spring 2022).
- Contributed to science popularization through public talks, posters, and gravitational-lensing demonstrations at IUCAA.
- Strong English proficiency; experienced in international teamwork, scientific presentations, and communicating science to specialist and general audiences.

Languages: English | Hindi | Marathi